UE26MA141A Engineering Mathematics – I · Unit 2 · Higher-Order Linear Differential Equations

Graphs and diagrams

Every graph and diagram in scope, redrawn as clean vector figures.

Class 014 · Applications of LDEs: LCR circuits · compiled 17 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
added for clarity	Standard context that is not in the class material
transcript only	Said in the lecture but not on any slide
video only	Seen only in a frame of the class video
not on current slides	From older material or past papers, not this year's slides
S014·8	Unit 2 deck 014 (“Applications of DE – LCR circuit”), slide 8 (rendered .pptx)
T014 00:04:29	Unit 2 AV-summary transcript “Applications of higher-order linear differential equations – RLC Circuits” (11.6 min), segment start
CW U2 P34	Unit 2 class-work problem 34
ESA Jan-24 s3 3b	Past paper: UE23MA141A Jan 2024 set 3, question 3.b

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1. Figure G1.1 · The series LCR circuit (slide 014·4, redrawn)
2. Figure G2.1 · Slide example: $Q(t)$ and $I(t)$ for $R = 8,\ L = 2,\ C = 0.1,\ E = 10$
3. Figure G2.2 · Damping regimes: R^2 against $\frac{4L}{C}$
4. Figure G2.3 · Slide 014·16: current for $E = 155 \sin 377t$, transient and steady state
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6. Figure G3.2 · CW P36: free discharge of the capacitor
7. Figure G4.1 · RL circuit (CW P34): current with its transient

Slide 014·4 has the only diagram (the series circuit), redrawn in G1.1; every other figure is added for clarity. Colours follow the Okabe–Ito order: blue = charge $Q(t)$ or the main response, vermilion = current $I(t)$ or a comparison, green = a third case; grey dashed lines are references (steady state, envelopes, CE). Every plotted curve is the sympy-verified solution of its worked example.

1 Modelling the series LCR circuit

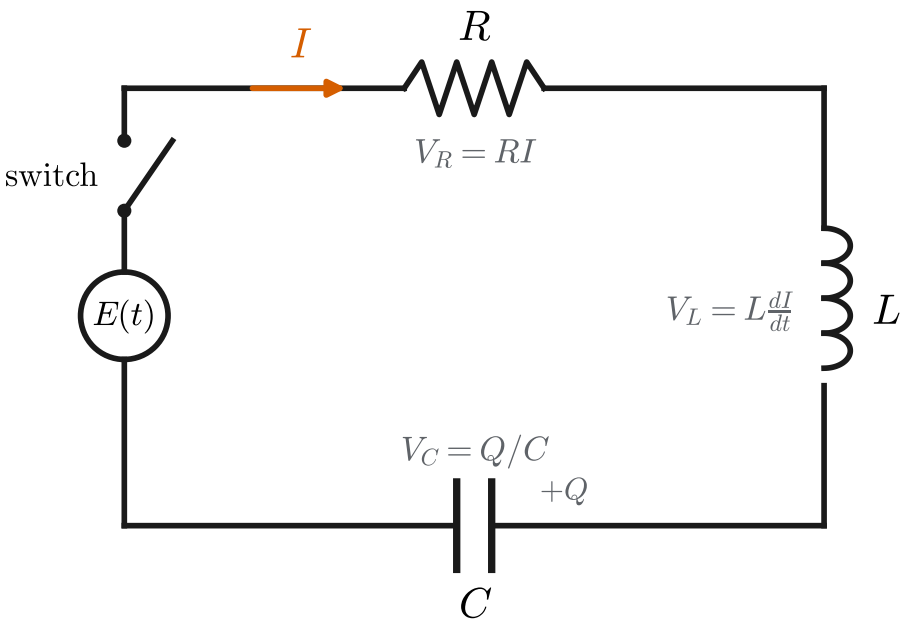


Figure G1.1. The series LCR circuit (slide 014·4, redrawn). ESA ★★

S014·4 · T014 00:00:13

How to read it. Source $E(t)$ and switch on the left, resistor R on top, inductor L on the right, capacitor C at the bottom, all in one loop so the same current I flows through each. The three drops $V_R = RI$, $V_L = L\frac{dI}{dt}$, $V_C = \frac{Q}{C}$ add up to $E(t)$. Schematic, not to scale.

2 Solving the circuit equation: CF, PI and damping

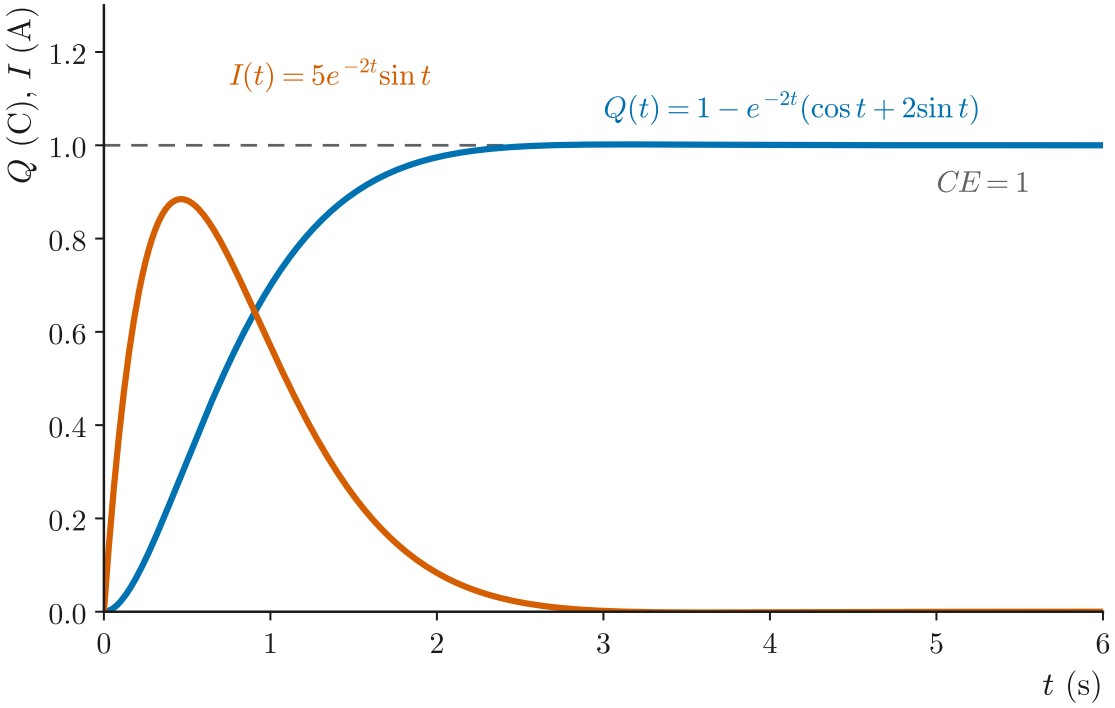


Figure G2.1. Slide example: $Q(t)$ and $I(t)$ for $R = 8$, $L = 2$, $C = 0.1$, $E = 10$. ESA ★★

ADDED FOR CLARITY S014·12

How to read it. The charge (blue) rises from 0 to the steady state $CE = 1$ C (grey dashed) with a small overshoot (underdamped, e^{-2t}). The current $5e^{-2t}\sin t$ (vermilion) starts at 0, peaks near $t = 0.46$ s and dies out: a battery drives no steady current through a capacitor.

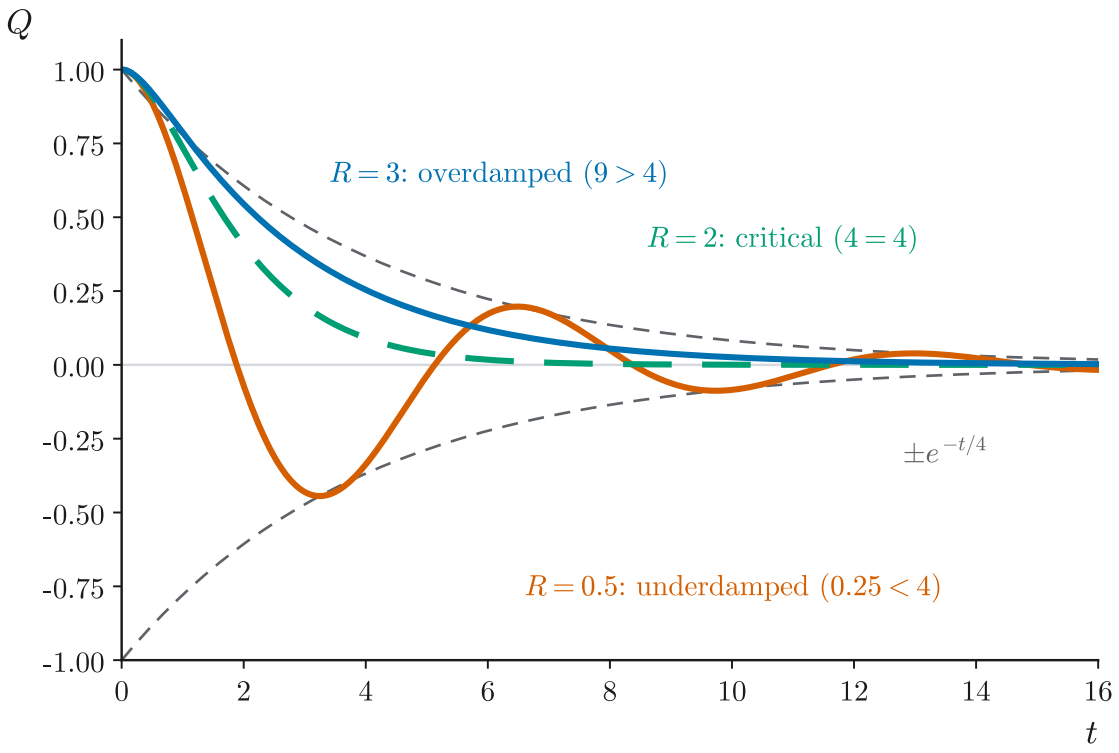


Figure G2.2. Damping regimes: R^2 against $\frac{4L}{C}$. ESA ★

ADDED FOR CLARITY S014·9 · S014·15

How to read it. Free discharge $Q'' + RQ' + Q = 0$ ($L = C = 1$, so $\frac{4L}{C} = 4$) from $Q(0) = 1$, $I(0) = 0$. $R = 3$ (blue, overdamped) creeps back slowly; $R = 2$ (green dashed, critical) returns fastest without crossing zero; $R = 0.5$ (vermilion, underdamped) oscillates inside the envelope $\pm e^{-t/4}$ (grey).

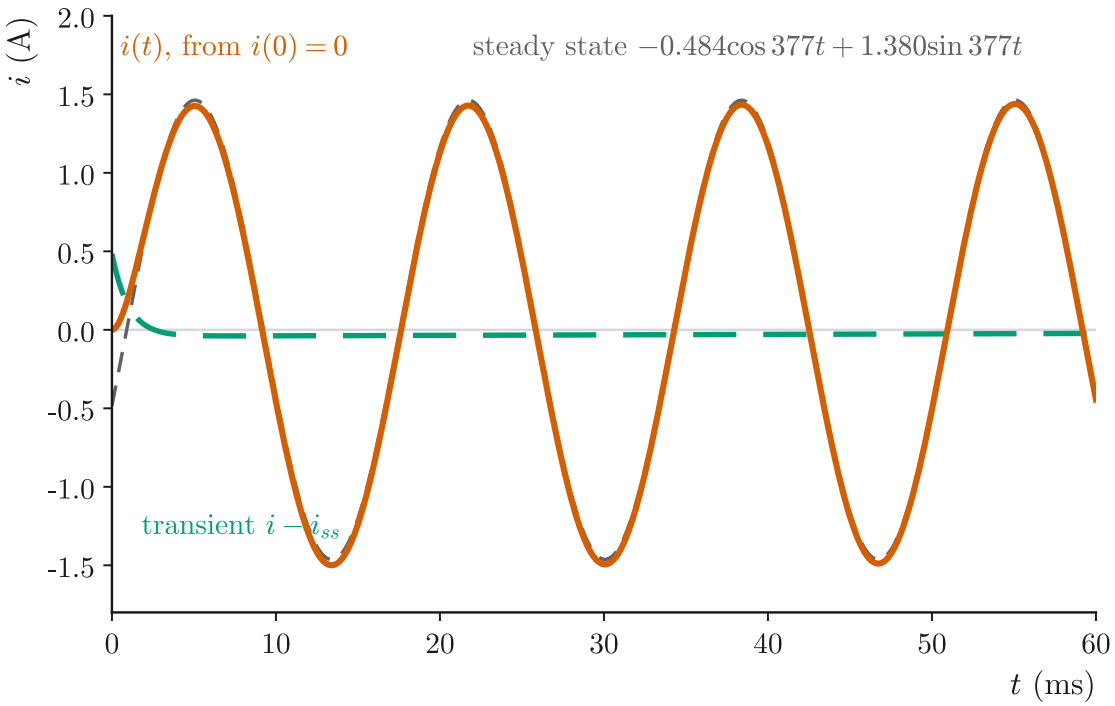


Figure G2.3. Slide 014·16: current for $E = 155 \sin 377t$, transient and steady state. ESA ★

ADDED FOR CLARITY S014·16

How to read it. The full current $i(t)$ (vermilion), the steady state $-0.4838 \cos 377t + 1.3804 \sin 377t$ (grey dashed) and their difference, the transient (green dashed). The fast part ($e^{-989.9t}$) is gone within a few milliseconds; the slow one ($-0.0424e^{-10.1t}$) shifts the curve slightly for a few tenths of a second.

3 Past-paper circuits: AC sources and free discharge

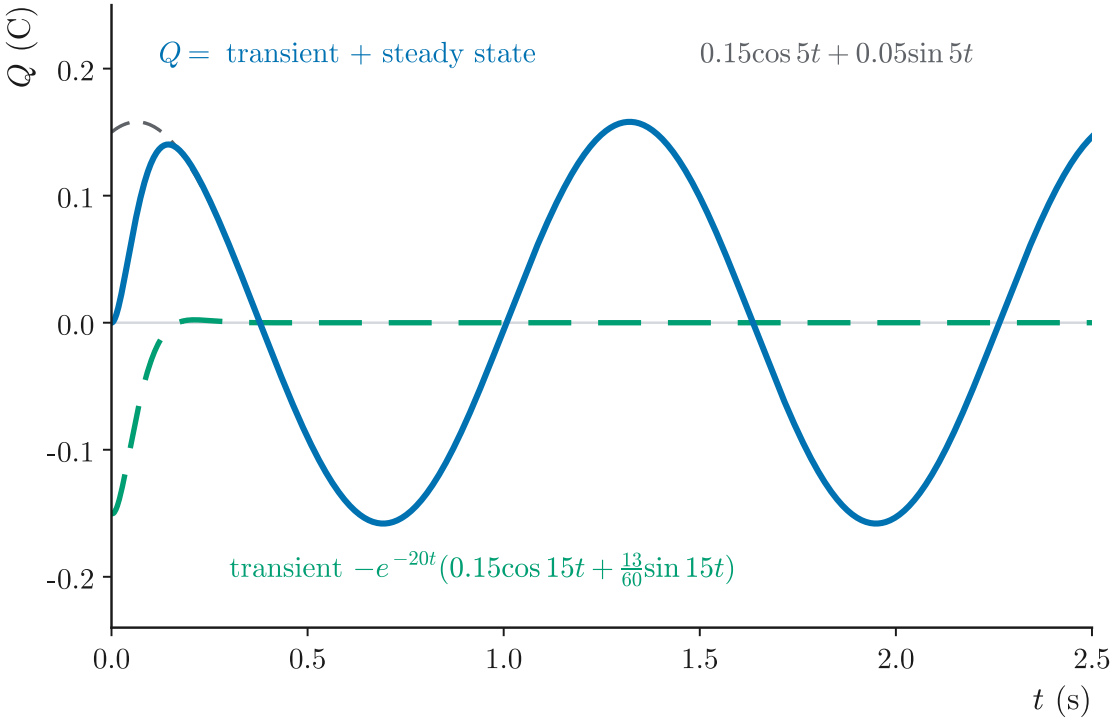


Figure G3.1. ESA $E = 100\cos 5t$ with $Q(0) = I(0) = 0$: charge = transient + steady state. ESA ★★

ADDED FOR CLARITY ESA Jan-24 s3 3b

How to read it. The charge $Q(t)$ (blue) is the sum of the transient $-e^{-20t}(0.15\cos 15t + \frac{13}{60}\sin 15t)$ (green dashed), which is gone by $t \approx 0.25$ s, and the steady state $0.15\cos 5t + 0.05\sin 5t$ (grey dashed, amplitude 0.158 C, period $\frac{2\pi}{5}$).

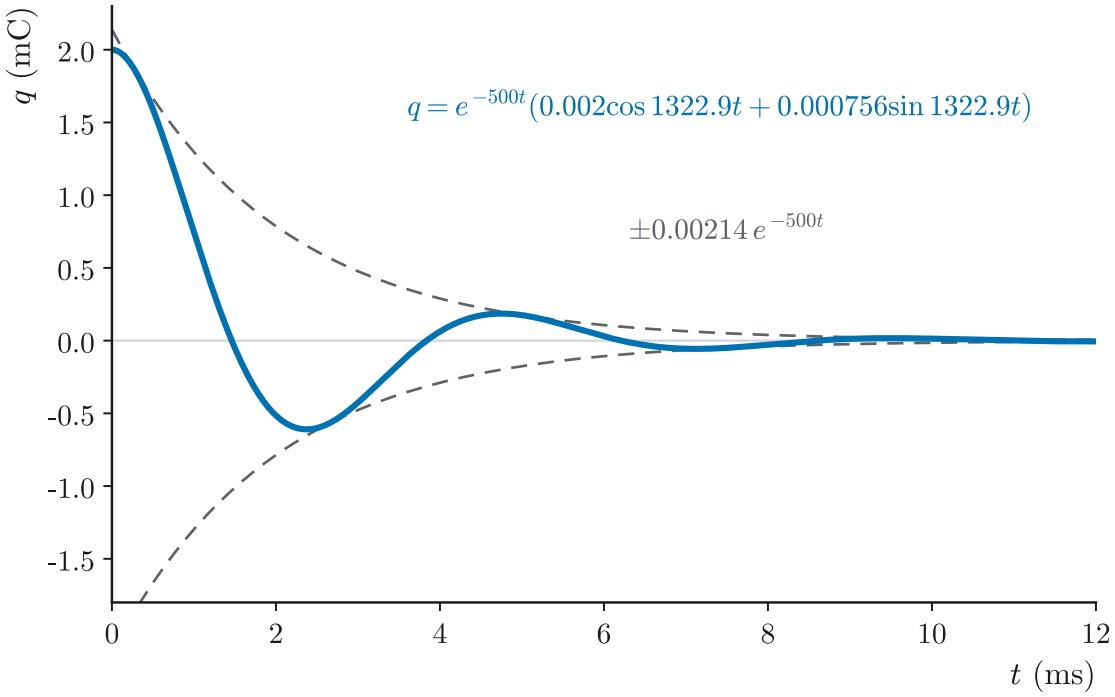


Figure G3.2. CW P36: free discharge of the capacitor. ESA ★★

ADDED FOR CLARITY CW U2 P36

How to read it. $q(t) = e^{-500t}(0.002\cos 1322.9t + 0.000756\sin 1322.9t)$ (blue) oscillates with period $\frac{2\pi}{1322.9} = 4.75$ ms inside the envelope $\pm 0.00214 e^{-500t}$ (grey dashed) and is negligible after about 10 ms.

4 First-order circuits: RL and RC

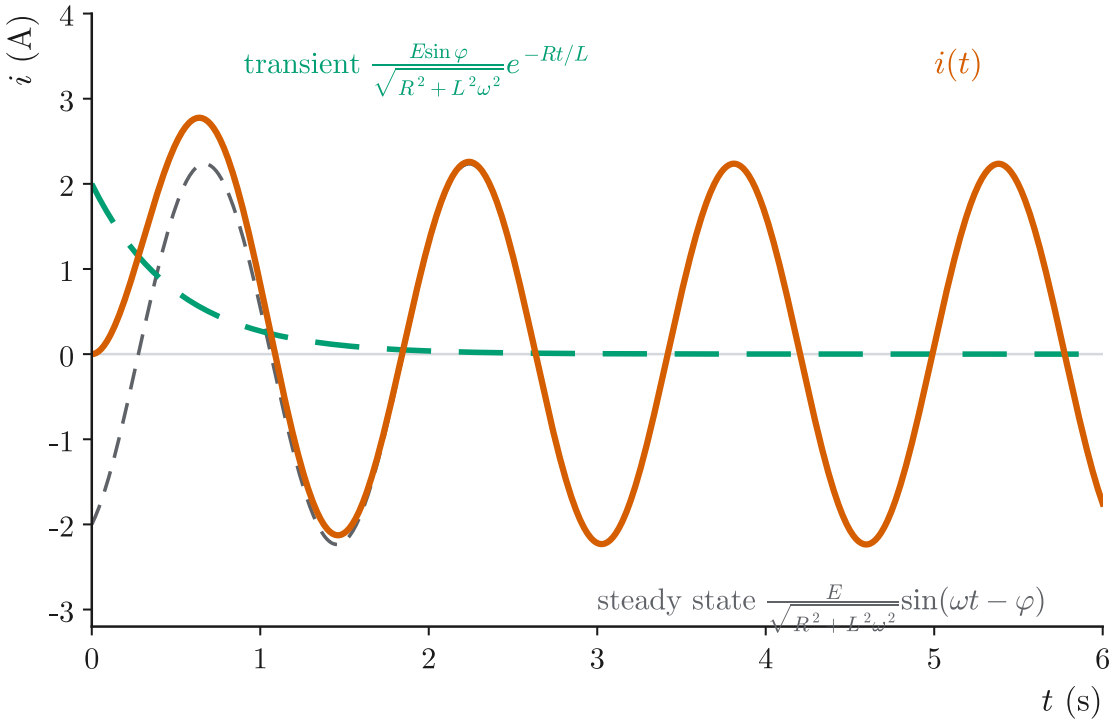


Figure G4.1. RL circuit (CW P34): current with its transient. ESA ★

ADDED FOR CLARITY CW U2 P34

How to read it. Illustrative values $E = 10$ V, $R = 2\ \Omega$, $L = 1$ H, $\omega = 4$ rad/s ($\varphi = \tan^{-1} 2$). The current (vermilion) starts at 0; the steady state $\frac{E}{\sqrt{R^2+L^2\omega^2}}\sin(\omega t-\varphi)$ (grey dashed) lags the source by φ; the difference is the transient $\frac{E\sin\varphi}{\sqrt{R^2+L^2\omega^2}}e^{-Rt/L}$ (green dashed), time constant $\frac{L}{R} = 0.5$ s.